# SHIMUL SHARMA

## BLOCKCHAIN / WEB3 DEVELOPER

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## PROFESSIONAL SUMMARY


Solidity Developer and Web3 backend engineer with hands-on experience building smart contracts, DAO governance systems, ERC-20/ERC-721 applications, DeFi protocols, and wallet/presale DApps. Strong in Foundry, Hardhat, Ethers.js, FastAPI/Fastify backends, Docker, GitHub Actions, fuzz/invariant testing, and smart contract security review.


## EDUCATION

**Bachelor of Technology - Electronics & Communication Engineering (VLSI)** *2024 - Expected 2028*
*Maharaja Agrasen Institute of Technology (MAIT), Delhi, India*

## TECHNICAL SKILLS

**Languages:** Solidity, TypeScript, JavaScript, Python, Go
**Smart Contracts & Standards:** ERC-20, ERC-721, UUPS Proxy, ERC-1967
**Blockchain & Integrations:** Ethereum, Binance Smart Chain, Chainlink VRF V2.5, Chainlink Automation, Ethers.js, Web3.js
**Frontend:** Next.js (App Router), React
**Backend & APIs:** Fastify, FastAPI, WebSocket Protocol, PostgreSQL, Redis
**Cryptography & Identity:** Ed25519 Signing, Merkle Trees (RFC 6962), W3C Verifiable Credentials, DIDs, JWT
**DevOps & Infrastructure:** Foundry, Hardhat, Forge, Anvil, Docker, Docker Compose, GitHub Actions (CI/CD), pnpm/Turborepo
**Security & Testing:** Slither, Aderyn, Fuzz Testing, Invariant Testing, Fork Testing, Access Control, Reentrancy, Oracle Staleness, CEI Pattern, Gas Optimization

## FREELANCE EXPERIENCE

**Freelance Solidity & Web3 Developer** *Dec 2025 - Feb 2026*
*Upwork - Remote*

- Engineered an ERC-20 token presale DApp with Solidity smart contracts and a Vite + React frontend.
- Integrated MetaMask and WalletConnect for wallet connectivity with real-time presale tracking using Ethers.js.
- Built a multi-chain DApp browser wallet supporting Ethereum (ERC-20) and Binance Smart Chain (BEP-20).
- Implemented encrypted private-key management via CryptoJS and Web3.js RPC integration compatible with Uniswap, PancakeSwap, and major DApps.

## PROJECTS

**DeFi Stablecoin Protocol (DSC) - *GitHub***

- Built a USD-pegged overcollateralized stablecoin backed by WETH/WBTC with liquidation incentives and health-factor checks.
- Integrated Chainlink price feeds with stale-oracle protection; applied ReentrancyGuard and CEI for safer state transitions.
- Wrote handler-based invariant/fuzz tests with 128,000 randomized calls and 0 reverts.

**FOUNDRY - Smart Contract Lottery - *GitHub***

- Architected an on-chain raffle using Chainlink Automation for scheduled upkeep and winner selection.
- Integrated Chainlink VRF V2.5 for tamper-proof randomness and authored a custom VRFCoordinatorV2_5Mock for Anvil tests.
- Configured GitHub Actions CI/CD for Anvil and Sepolia deployment with Etherscan verification.

**NFT ERC-721 Project - *GitHub***

- Built Foundry-based ERC-721 NFT contracts for IPFS-hosted metadata and fully on-chain SVG NFTs.
- Implemented deployment scripts, local/testnet flows, Base64 metadata encoding, gas snapshots, and Foundry unit/fork testing.

**Decentralized Autonomous Organisation (DAO) - *GitHub***

- Built an on-chain governance system with OpenZeppelin Governor, ERC20Votes, ERC20Permit, and TimelockController.
- Implemented proposal, vote, queue, and execute lifecycle controlling a target Box contract through the DAO timelock.
- Added Foundry end-to-end governance tests and stateful invariant/fuzz tests covering ownership, roles, token supply, and execution rules.

**PROVORA - Verifiable Exam Integrity Platform - *GitHub***

- Built and deployed a full-stack exam-integrity platform (Next.js, Fastify/TypeScript API, Python FastAPI scoring service) generating explainable, evidence-backed flags instead of opaque "suspicious activity" verdicts.
- Implemented passwordless DID login (Ed25519 challenge-response) and Ed25519-signed W3C Verifiable Credentials for portable, self-owned exam credentials.
- Built a hash-chained Merkle transparency log (RFC 6962) and a Go/WebSocket/Redis realtime service for live behavioral event streaming.

## SECURITY PORTFOLIO

**Smart Contract Security Audit Reports - *GitHub***

Includes Solidity audit reports and findings covering access control issues, on-chain data exposure, reentrancy/state manipulation risks, logic errors, gas optimizations, and mitigation recommendations.